JAVA CORE PROGRAMMING ASSESSMENT

Topics: Data Types, Variables, Operators, Control Flow, Classes, Objects, Constructors, Encapsulation, Scanner

Name: _____ **Date:** _____ **Score:** _____ / 40

1. Which data type is used to store fractional numbers in Java by default?

A) float
B) double
C) int
D) long

2. What is the default value of an uninitialized instance boolean variable in Java?

A) true
B) false
C) null
D) 0

3. Which of the following is an invalid Java variable name?

A) _score
B) $amount
C) 2ndValue
D) user_name

4. What is the output of the expression `10 % 3` in Java?

A) 3
B) 1
C) 3.33
D) 0

5. What will be printed by the following snippet?

```
int a = 5; System.out.println(a++ + ++a);
```

A) 10
B) 11
C) 12
D) 13

6. Which operator is used to compare two primitives for equality?

A) =
B) ==
C) .equals()
D) ===

7. What is the result of `5 > 3 && 8 < 4`?

A) true
B) false
C) Compilation Error
D) Runtime Exception

8. Which Scanner method reads a single full line of text including spaces?

A) next()
B) nextLine()
C) readLine()
D) getString()

9. To use the `Scanner` class, which package must be imported?

A) java.lang
B) java.io
C) java.util
D) java.net

10. How do you instantiate a Scanner to read input from the console?

```
Scanner sc = new Scanner(_______);
```

A) System.in
B) System.out
C) File.input
D) Console.in

11. What happens if Scanner's `nextInt()` encounters a non-integer token?

A) Returns 0
B) Throws InputMismatchException
C) Casts value automatically
D) Compilation Error

12. Which statement regarding an `if-else` construct is true?

A) The else block is mandatory.
B) Condition must evaluate to a boolean.
C) Integers can be used directly as conditions.
D) Multiple else blocks are allowed.

13. What is the output of the following block?

```
int x = 10; if (x = 5)
System.out.println("Yes"); else
System.out.println("No");
```

A) Yes
B) No
C) Compilation Error
D) Runtime Error

14. Which of the following data types CANNOT be used in a Java `switch` statement?

A) int
B) String
C) double
D) char

15. What occurs if a `break` statement is omitted in a matching switch case?

A) Compilation fails immediately
B) Execution falls through to subsequent cases
C) The program enters an infinite loop
D) Default case executes automatically

16. What will be printed by this switch statement?

```
int ch = 2; switch(ch) { case 1:
System.out.print("A"); case 2:
System.out.print("B"); case 3:
System.out.print("C"); break; default:
System.out.print("D"); }
```

A) B
B) BC
C) BCD
D) ABC

17. A Class in Object-Oriented Programming is best described as:

A) An instance of an Object
B) A blueprint or template for objects
C) A primitive data structure
D) A memory location holding values

18. Which keyword is used to create an instance of a class in Java?

A) alloc
B) create
C) new
D) class

19. Where are object instances stored in Java memory architecture?

A) Stack
B) Heap
C) Register
D) Code Segment

20. What is the default initial value of an object reference field in a class?

A) 0
B) "" (empty string)
C) null
D) undefined

21. What is the primary role of a Constructor in Java?

A) To destroy unreferenced objects
B) To initialize a newly created object
C) To execute static blocks
D) To compile class code

22. Which of the following is TRUE about Java constructors?

A) They must have a return type of void.
B) Their name must match the class name exactly.
C) They can be declared as static.
D) They cannot take any arguments.

23. When does Java compiler provide a default no-arg constructor?

A) Always for every class created
B) Only when no user constructors are defined
C) Only when class is declared abstract
D) Whenever class contains instance variables

24. What is Constructor Overloading?

A) Creating constructors in subclass
B) Multiple constructors with different parameters
C) Calling a constructor infinitely
D) Declaring constructors with different return types

25. Which keyword is used inside a constructor to invoke another constructor of the same class?

A) super()
B) this()
C) self()
D) main()

26. Encapsulation is defined as:

A) Inheriting properties from a parent class
B) Wrapping data and methods into a single unit
C) Overloading methods with identical names
D) Converting primitives into wrapper objects

27. How can field encapsulation be achieved in Java?

A) Declare variables public, provide public getters
B) Declare variables private, provide public getters/setters
C) Declare variables protected with no methods
D) Declare variables static private only

28. Which access modifier hides class members from all external classes?

A) public
B) protected
C) default
D) private

29. What is a major practical benefit of Encapsulation?

A) Faster program execution speed
B) Control over data read/write validation
C) Automatic memory cleanup
D) Eliminates need for constructors

30. What is the conventional naming standard for a getter method of a boolean property `active`?

A) getActive()
B) isActive()
C) fetchActive()
D) booleanActive()

31. What size in memory does a standard `int` occupy in Java?

A) 2 bytes **B)** 4 bytes
C) 8 bytes **D)** Depends on OS

32. Which explicit cast is valid without precision loss risk?

A) double to int **B)** int to double
C) long to int **D)** float to long

33. What will be stored in `x` after executing `int x = (int) 7.99;`?

A) 7.99 **B)** 8
C) 7 **D)** Compilation Error

34. Which scope limits a variable to the method block in which it is declared?

A) Instance **B)** Global
C) Local **D)** Static

35. What does the `this` keyword refer to in Java instance methods?

A) The current class definition
B) The current object instance
C) The parent superclass object
D) The main executing thread

36. What happens if you define a method with the exact same name as the class?

A) It becomes a constructor.
B) It is a regular method if return type is stated.
C) Causes a compile error immediately.
D) Overrides default object constructor.

37. Which short-circuit operator executes right operand ONLY if left is false?

A) && **B)** ||
C) ! **D)** ^

38. What is the output of: `System.out.println("A" + 1 + 2);`?

A) A3 **B)** A12
C) 3A **D)** Error

39. How do you close a Scanner instance named `sc` to release system resources?

A) sc.delete() **B)** sc.close()
C) sc.stop() **D)** sc.free()

40. Which conditional ternary expression is equivalent to `if(a > b) max = a; else max = b;`?

A) max = (a > b) ? a : b;
B) max = (a > b) : a ? b;
C) max = (a < b) ? a : b;
D) max = if(a > b) ? a : b;

ANSWER KEY

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	**9**	C	**17**	B	**25**	B	**33**	C
2	B	**10**	A	**18**	C	**26**	B	**34**	C
3	C	**11**	B	**19**	B	**27**	B	**35**	B
4	B	**12**	B	**20**	C	**28**	D	**36**	B
5	C	**13**	C	**21**	B	**29**	B	**37**	B
6	B	**14**	C	**22**	B	**30**	B	**38**	B
7	B	**15**	B	**23**	B	**31**	B	**39**	B
8	B	**16**	B	**24**	B	**32**	B	**40**	A